ScienceVision: Can AI Observe and Reason Like a Scientist?

UniPat AI, Elorian AI

Abstract

Scientific discovery begins with observation, and frontier models are increasingly used to accelerate this process across scientific disciplines. Yet many scientific questions require models to understand evidence beyond text and 2D images, such as spatial contacts in protein structures and spatial relationships in 3D anatomy. Many multimodal benchmarks evaluate reasoning over text and 2D images, while scientific 3D benchmarks differ in disciplinary scope, representations, and protocols, making cross-disciplinary comparison difficult. To address this gap, we introduce **ScienceVision**, a cross-domain benchmark for scientific 3D understanding and reasoning through expert-level visual question answering (VQA). It contains 250 questions, with 50 each from biology, chemistry, medical imaging, mechanical CAD, and architecture. Each task pairs a native 3D artifact with a scientific question, a specified response format, and a task-specific reference answer, requiring domain knowledge to be applied to geometric or topological evidence. We evaluate 12 models on the same questions under three explicit evidence-access conditions: programmatic analysis of source artifacts, the same access augmented with expert-inspired inspection and measurement tools through the Model Context Protocol (MCP), and fixed six-view renderings as an image-based control. The inspection environment supports view changes, target isolation, and slice inspection. In this main comparison, the best mean answer score is 0.53 out of 1. Source access outperforms fixed views for every model, by 17.2 percentage points on average. Additional MCP tools are associated with smaller score changes for every model, with no consistent direction. Measured performance is thus more sensitive to access to native 3D evidence than to additional inspection tools, making these access conditions essential to interpreting cross-disciplinary 3D understanding and reasoning scores.

HOMEPAGE: `https://github.com/UniPat-AI/ScienceVision`

1 Introduction

Scientific observation begins with a question about an object and the evidence needed to answer it. Identifying atoms hidden in an initial molecular view can determine which bonded path must be analyzed. Locating a lesion in a CT volume requires interpreting anatomy and returning a location in the specified coordinate frame. The task is to connect evidence from the supplied object to a domain-specific conclusion.

For an AI system, this connection can involve selecting a target, revealing a structure, measuring a relation, and applying scientific definitions or conventions. The relevance of a measurement depends on which object was identified and what the question asks. A correct distance between the wrong atoms does not answer a molecular-contact question. Likewise, a plausible anatomical description does not supply a requested voxel location.

Existing evaluations already study scientific structural question answering and agent interaction with native 3D data. DeepTumorVQA evaluates questions about CT volumes, and SciVisAgentBench includes scientific analysis, visualization, and question-answering outcomes across domains (Chen et al., 2026; Ai et al., 2026). ScienceVision centers the evaluation unit on a scientific question about a supplied artifact and a checkable answer. It places five domains under a shared task and reporting framework, allowing evidence-access conditions to be compared while retaining their execution limits.

ScienceVision contains 250 questions across architecture, biology, chemistry, mechanical CAD, and medical imaging, with 50 questions per domain and 22 subdomains. The questions combine structural evidence with domain concepts, object identity, or explicit conventions. Expert-level describes the intended knowledge required to interpret a task; it does not denote measured difficulty relative to a human baseline.

The benchmark supports programmatic analysis of the source, source analysis with inspection tools exposed through MCP, and a fixed-image condition. Source access permits computations from geometry or other artifact data. The MCP condition adds viewer operations such as changing views, isolating structures, navigating slices, and measuring geometry. These are access conditions for evaluating existing systems; the interface is not presented as a new reasoning algorithm.

Our contributions are:

- A benchmark of scientific questions about supplied 3D artifacts, with task-specific answer contracts across five equally represented domains.
- Task, reference, and inspection resources that document the intended evidence and support checking submitted answers, with reference provenance distinguished from runtime validation.
- An empirical assessment of 12 models in three evidence-access conditions, with available-repeat coverage and known harness provenance reported alongside the comparison limits and the limits of attributing score differences to observation, reasoning, or tool access.

2 Related Work

2.1 Scientific structural question answering

DeepTumorVQA organizes questions about 3D CT into recognition, measurement, visual reasoning, and medical reasoning, with reference answers generated by deterministic programs from validated imaging metadata (Chen et al., 2026). It reports multiple-choice answer accuracy and rule-based scoring of free-form responses. ProtStructQA defines questions over AlphaFold-predicted proteins through a hidden typed domain-specific language, covering structural measurements and labels (Mandiga et al., 2026). Program execution supplies each reference value, and predictions are checked using answer-type-specific tolerances, exact matching, or set overlap. Both benchmarks connect object-derived evidence to explicit answer criteria within a scientific domain. *ScienceVision* uses task-specific references and answer contracts across five scientific and engineering domains, specifying requested quantities, identities, and locations under the relevant domain conventions. Table 1 compares scientific scope, question-answering objectives, evidence access, and formal scoring across related benchmarks.

Table 1: Comparison of benchmark scope, question-answering (QA), evidence access, and grading. ✓: reported; ×: not reported. Cross-domain denotes multiple scientific fields. QA focus includes captioning classified as QA by the benchmark authors; Partial denotes QA alongside visualization or processing tasks. Interaction includes agent-selected views, slices, crops, or rendered prediction feedback. Static-image QA denotes a separate fixed-image evaluation. Deterministic scoring considers formal outcomes: ✓, rules without LLM judges; Partial, mixed rule-based/LLM grading; ×, LLM-only grading.

Benchmark	**Cross-domain science**	**QA focus**	**Visual interaction**	**Static-image QA**	**Deterministic scoring**
DeepTumorVQA (Chen et al., 2026)	×	✓	✓	✓	✓
ProtStructQA (Mandiga et al., 2026)	×	✓	×	×	✓
OmniCAD (Wang et al., 2026)	×	×	✓	×	✓
SciVisAgentBench (Ai et al., 2026)	✓	Partial	✓	×	Partial
GLUE3D (Mariani et al., 2026)	×	✓	×	✓	Partial
***ScienceVision* (ours)**	✓	✓	✓	✓	✓

2.2 Scientific visualization agents and evidence access

SciVisAgentBench evaluates scientific analysis and visualization workflows in established software, including analytical question answering and spatial outputs (Ai et al., 2026). It grades produced outcomes against expert-authored rubrics using multimodal judges and deterministic evaluators. *ScienceVision* centers its task families on scientific questions about supplied artifacts and grades submitted answers against task-specific references using deterministic checks without an LLM judge. The distinction concerns the task objectives and graded outcomes; question answering and interaction are already present in prior scientific 3D evaluations.

Scientific agents obtain evidence through generated analysis scripts, visual feedback, and direct viewer operations (Peterka et al., 2025; Liu et al., 2024; 2025). MCP interfaces also expose operations for molecular visualization, bioimages, and medical volumes (egtai, 2025; Lawrence Livermore National Laboratory, 2025; zhaoyouj, 2025). *ScienceVision* uses these access mechanisms as evaluation conditions: source data with code execution, the same access with additional MCP inspection tools, and fixed rendered views. These conditions distinguish the evidence a solver receives and the operations it may request from the final answer being graded. Tool availability alone does not establish how a particular answer was obtained.

3 The ScienceVision Benchmark

ScienceVision asks for structural and spatial judgments about specific 3D scientific objects. Its 250 tasks are evenly distributed across architecture, biology, chemistry, mechanical CAD, and medical imaging (Figure 1). The questions concern properties of particular objects, such as continuity across storeys in a building, contact between macromolecular structures, or the shape and multiplicity of assembly components.

3.1 Objects, targets, and answer contracts

Each task pairs an object with a question, an answer contract, and a reference answer. The question can identify its target by a scientific description or by its appearance in a specified view. The contract defines what to report and fixes the relevant units, coordinates, and object-identity conventions. Responses range from a single category to several related quantities or locations. Expert inspection records preserve the viewer operations and visual states associated with the reference annotations; Appendix A details both the records and answer formats.

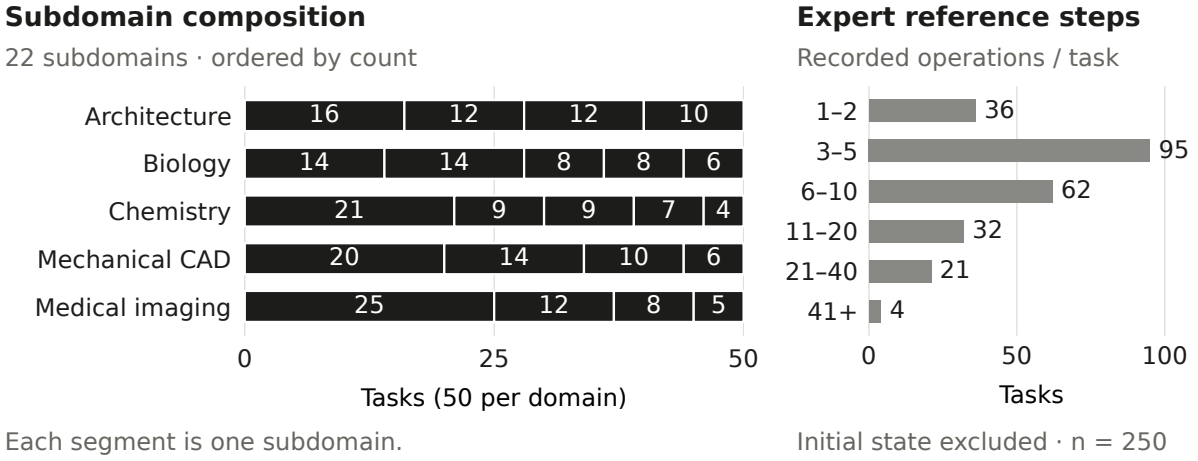
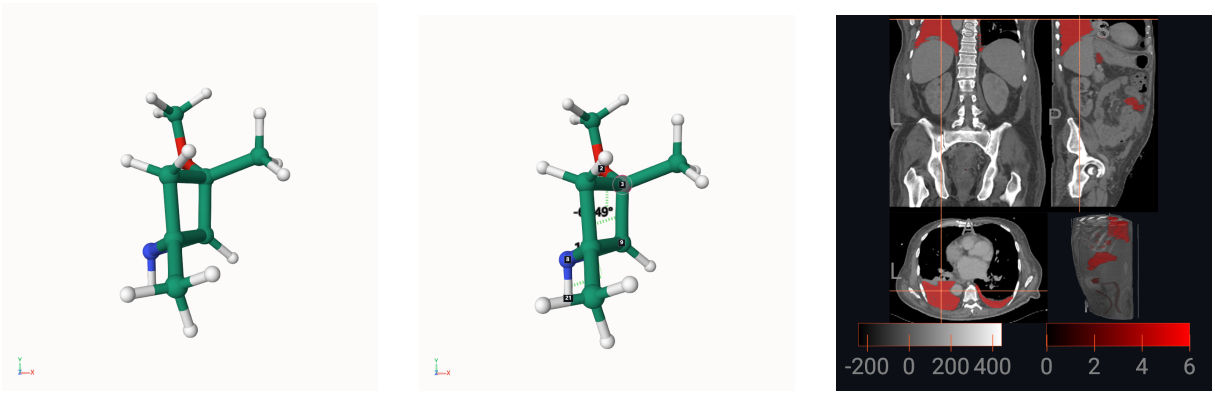


Figure 1: Subdomain composition (left; names in Appendix A) and expert reference inspection lengths (right).

3.2 From target selection to scientific judgment

Chemistry: a view-defined target and its bonded geometry. In task `sv-chem-chem_conformation-0002` (Figure 2a,b), the initial camera view defines two atoms by their occlusion. Their identities select the endpoints of a shortest bonded path. The task requests the intermediate atom IDs, excluding both endpoints, and the smallest absolute dihedral angle over consecutive four-atom segments along the path. The stored reference gives 64.49° for this path minimum.

Medical imaging: a named lesion and its spatial extent. Task `sv-medi-med_ct_geometry-0103` supplies the diagnostic target, a small left pleural effusion in a CT volume. It asks for the lesion's extent along the three array axes, every longest axis, and a signed left–right centroid offset. The reference extent is $[86, 81, 61]$ voxels. Extents include both bounding voxels, whereas the centroid offset is a physical distance in millimetres.



(a) Chemistry: initial view.

(b) Chemistry: labels and measurements.

(c) Medical: CT inspection.

Figure 2: Original recorded inspection states for the chemistry (a,b) and medical imaging (c) examples.

4 Construction and Reference Verification

4.1 Question construction and expert annotation

Asset checks establish whether the assets support the task, are paired correctly when needed, and can be inspected in the viewer. Candidate questions are prepared from templates or curated drafts, with optional agent assistance. Candidates are checked against task rules, selected deterministically, and packaged with their assets for expert review.

Experts check the questions' targets and domain assumptions and edit them as needed. They inspect the 3D evidence to provide reference answers. Available inspection records document how the answers were obtained. Quality review determines which records can be accepted and which require human or agent revision.

4.2 Reference answers and recorded inspections

A reference answer, a recorded inspection, an oracle program, and a scoring comparator serve different purposes. The reference specifies the intended target. A reference inspection records a route used to inspect the object. An oracle program may derive a value from geometry, invoke tools, or return a recorded reference. The comparator decides whether a submitted answer satisfies the task's accepted-answer rules.

The current frozen release classifies 110 references as recorded values, 106 as script-derived values, and 34 as tool-derived values. This is a provenance classification, not a success rate for re-executing the oracles. In particular, the recorded-reference class must not be described as uniformly recomputable from the released assets. The release audit did not execute all reference programs or runtime scorers.

Recorded inspections make the intended work inspectable, but their actions are not a measured minimum and need not match an agent's successful route. Initial frames, screenshots, recorded actions, model trajectory steps, and shell/tool calls are different units. None supplies a human task-success baseline.

4.3 Verification and accepted answers

Task-level verification must establish which answers are accepted, how units and coordinates are interpreted, and how equivalent representations are handled. A lesion-pointing question can admit multiple points; an object identity can have accepted aliases; numerical judgments can require task-specific tolerances. A single displayed reference does not define these rules.

The current configuration inventory contains 54 task-specific overrides. It distinguishes explicit component rubrics, accepted aliases, and tolerance overrides; these categories overlap. This inventory describes

configuration, and does not certify the behavior of the external default scorer or each historical run. The scoring unit and aggregation are defined in Section 5.

5 Evaluation Protocol and Experimental Setup

5.1 Evidence-access conditions

Source supplies the native artifact and code execution. The agent can inspect source data and compute quantities such as distances or structural relations. **Source+MCP** adds viewer inspection and measurement tools to the same source access. Tool use is optional, so this condition evaluates a combined workflow rather than observation exclusively through a viewer.

Fixed6 supplies the question and six pre-rendered views in place of the artifact. The camera states and image order are fixed before the run; the agent cannot request another view. Medical tasks pair volume renders with slices, so six nominal views can contain twelve image files. Appendix C collects interface details and remaining documentation gaps, including the limits of historical access enforcement. The earlier aligned/orbit sweep uses a separate task version and is reported as a historical design probe.

5.2 Models and execution configurations

We evaluate the 12 models listed in Table 7 on 250 tasks, with 50 tasks in each of five domains, under the three access conditions. An execution configuration combines a language model, an agent harness, and an evidence-access condition. The harness manages the model's interaction with files, code execution, and tools; Codex and Claude Code denote this execution layer, not additional models or access conditions.

We use Codex for GPT-6 Astra and GPT-6.1 Sol and Claude Code for the other ten models across all three conditions (OpenAI, 2026; Anthropic, 2026). These selected configurations define the main comparison. GPT Codex uses reasoning effort `max`; the Claude Code effort setting is not recorded. The Fixed6 harness assignment is documented at study level; its score summary lacks per-row harness metadata. Appendix D distinguishes configuration declarations from verified run metadata.

The nominal protocol uses three repeats per task; Gemini under Source and Source+MCP and Hunyuan under Fixed6 have incomplete repeat coverage, detailed in Appendix D. Supplementary cross-harness runs, including `high`-effort Fable/Opus Codex, are analyzed separately in Section 6.3.1.

5.3 Task scores and aggregation

For task i and repeat r, the submitted answer $\hat{y}_{ir}$ is compared with reference y_i^* using the task-specific rubric ρ_i:

$$s_{ir} = \mathrm{Eval}_i(\hat{y}_{ir}, y_i^*; \rho_i) \in [0, 1]. \tag{1}$$

The task-specific rubric specifies scored components, matching rules, and partial credit. Detailed scoring specifications are given in Appendix B.

For a fixed execution configuration, let R_i contain the available scored repeats. We first average within each task, then weight tasks equally:

$$\bar{s}_i = \frac{1}{|R_i|} \sum_{r \in R_i} s_{ir}, \qquad S_D = \frac{1}{|D|} \sum_{i \in D} \bar{s}_i. \tag{2}$$

Here $|D| = 50$ for a domain and $|D| = 250$ for the full benchmark. We retain exported repeat scores, omit unavailable repeats, and use no intermediate rounding. The reported mean task score permits partial credit and is distinct from complete-answer accuracy.

5.4 Comparisons and uncertainty

For Source+MCP minus Source, we resample paired task means over the 250 task IDs with replacement 2,000 times and report pointwise 95% percentile intervals. The resampling preserves each task's available-repeat mean. These intervals describe task-sampling uncertainty conditional on the evaluated outputs; they do not estimate repeated-run variability. Appendix E gives the seed, quantile calculation, and complete intervals.

The main table compares evaluated configurations. Source versus Fixed6 changes both the evidence representation and permitted computation, and the available records do not establish matched harness, model-version, input-build, and scorer settings for every retained run. A controlled estimate of MCP access would also require matching those factors and execution budgets. We therefore interpret score differences as configuration comparisons, without attributing them solely to active observation, tool use, or the underlying model.

6 Results and Analysis

6.1 Scores by model and access setting

Figure 3 compares the 12 selected model–harness configurations under Source and Source+MCP access. Appendix E.1 gives the complete three-setting scores and execution declarations (Table 7). Within these released leaderboard rows, GPT-6 Astra/Codex has the highest Source score, 0.5291; Claude Fable 5.1 has the highest Fixed6 score, 0.2965. Both source-access means exceed Fixed6 for every model (Table 7). The Source–Fixed6 gaps are 7.73–28.49 percentage points.

The rankings characterize these canonical model–harness configurations and can change under alternative execution settings.

These are mean task scores with partial credit, rather than fractions of completely correct answers. The differences concern evaluated configurations: source access permits direct calculations, and the cross-condition harness provenance is incomplete. They do not isolate the benefit of active visual inspection. Supplemental Claude Opus 5.5/Codex Source (0.5448) and Claude Fable 5.1/Codex Source+MCP (0.5349) scores exceed the selected table maxima. Section 6.3.1 reports these separate source layers without replacing the released one-harness-per-model comparison.

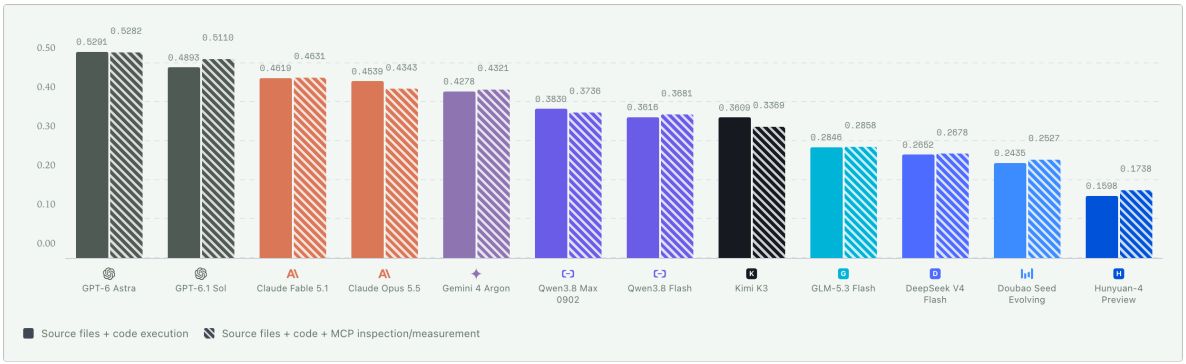


Figure 3: Source and Source+MCP leaderboard for the 12 canonical configurations, re-exported from the local ScienceVision blog source with unrounded task means. Solid bars show native source files and code execution; hatched bars add MCP inspection and measurement tools. Full model names and four-decimal scores are shown directly; colors identify vendors. These configurations do not include the supplementary cross-harness runs. Complete scores for all three settings are in Appendix E.1.

6.2 How does evidence access change the reported score?

Adding MCP changes the overall mean in both directions: eight configurations increase and four decrease. The paired differences range from -2.40 percentage points for Kimi K3 to $+2.17$ points for GPT-6.1 Sol. Astra's difference is -0.08 points, with a pointwise 95% task-bootstrap interval of $[-1.94, 1.83]$.

Only Sol's interval, $[0.34, 4.19]$, and Opus's interval, $[-3.83, -0.23]$, exclude zero among the 12 pointwise intervals. These intervals use no multiple-comparison adjustment and hold the evaluated outputs fixed. They describe task resampling, rather than repeated-run uncertainty or a controlled intervention on MCP access. Appendix E gives the complete paired estimates and resampling procedure.

6.3 Ablation studies and design probes

We examine sensitivity to the model–harness configuration and use two historical design probes to study fixed-view budgets and tool-only access. The historical probes do not estimate matched effects on the current release.

6.3.1 Execution harness and reasoning budget

Scores belong to model–harness configurations. We disclose the alternative harness runs under both Source and Source+MCP access so that the main table's author-selected vendor workflow is visible rather than hidden in a model name. GPT Codex uses `max`; the authors report using `high` for Fable/Opus Codex because `max` API calls encountered timeouts. The leaderboard retains the canonical configurations defined in Section 5, instead of mixing the highest values from heterogeneous exports.

The supplementary scores could change the apparent ranking: the separate Opus/Codex `high` batch scores 0.5448 under Source, above the canonical table's highest Source score of 0.5291. Its build is not fully matched to the release. In Table 2, Codex is higher in all eight 250-task comparisons by 4.56–13.40 percentage points. On the same 94 fix-affected IDs at reported high effort, Claude Code is higher by 1.81–4.51 points (Table 3). This purpose-selected subset is neither random nor domain-balanced; its means weight tasks equally. Matching IDs does not establish identical questions, targets, references, scorers or input builds. These results establish sensitivity to the evaluated configuration and evidence slice, not a universal ordering of harnesses. Appendix D.2 records the source layers and their build or effort differences.

Extra interaction can support error recovery; length alone is insufficient. In one archived Fable/Codex trial, a failed CIF-library import was followed by inspection of atom-table metadata, a replacement parser, and a distance calculation that received full credit under the archived rubric. This is a concrete use of additional code-feedback iterations. In the 3,000-trial Fable/Opus Codex `high` batch, however, full-score trials have a median of six agent response entries, compared with nine for zero-score trials; 47 trials end in an executor timeout. Effective budgets must support useful computation and timely completion. These observations do not identify a benefit from length itself, and completion tokens do not measure the unavailable reasoning-token count.

Tool orchestration is part of the execution setting. The inspected records show shell-execution calls for Codex and separate Bash, Read and Edit calls for Claude Code. These primitives express file inspection and computation at different granularities and permit different batching. The recovery example illustrates an inspect–parse–measure sequence in which intermediate feedback determines the next code operation. Attributing a score gain to one harness's ordering requires matched trajectories from both systems. Appendix D.3 gives the case and counting definitions. Appendix F gives additional provenance and diagnostics for the historical camera and MCP-only studies below.

Table 2: Cross-harness performance on 250 tasks.

Model	Access	Codex	Claude Code	Δ (pp)
GPT-6 Astra	Source	0.5291^{S2}	0.4082^{S3}	+12.09
	Source+MCP	0.5282^{S2}	0.3942^{S3}	+13.40
GPT-6.1 Sol	Source	0.4893^{S2}	0.3865^{S3}	+10.28
	Source+MCP	0.5110^{S2}	0.3843^{S3}	+12.67
Claude Fable 5.1	Source	0.5075^{S5}	0.4619^{S2}	+4.56
	Source+MCP	0.5349^{S5}	0.4631^{S2}	+7.18
Claude Opus 5.5	Source	0.5448^{S5}	0.4539^{S2}	+9.10
	Source+MCP	0.5322^{S5}	0.4343^{S2}	+9.78

Δ: Codex minus Claude Code, in percentage points. Source supplies source data and code execution; Source+MCP adds inspection tools. The route identifiers `cc`/`ccmcp` denote these access conditions; `cc` does not identify the harness. **S2:** current score release `59dae5a6`, archived outputs plus 39 task reruns; GPT Codex effort is max, and canonical Claude Code effort is unestablished. **S3:** GPT Claude Code outputs frozen on 2026-10-17, using the older question, gold and scorer layer without the final 39 reruns; effort is unestablished. **S5:** Fable/Opus Codex FO outputs dated 2026-10-20, high effort; high was used because max API calls encountered timeouts. GPT Codex uses max; these effort settings are not equalized budgets. Run-tree hash equality to the final task bundle is unverified. All three layers contain the same 250 IDs. Every S5 model–access cell retains three exported zero scores for one medical question across its three repeats, flagged ineligible; all 250 tasks remain in the denominator. We do not reapply an eligibility gate. These comparisons do not isolate a causal harness effect.

Table 3: The harness ordering reverses in the Fable/Opus 94-ID slice at reported high effort. This fix-selected subset is neither random nor domain-balanced. Each cell averages three scores per task and then the 94 tasks equally, rather than averaging domains equally. $\Delta(\text{pp}) = 100 \times (\text{Codex} - \text{Claude Code})$. The four observed differences favor Claude Code; task, reference, scorer and input-build equality remain unverified.

Model	Access	Codex (S5)	Claude Code (S4)	Δ (pp)
Claude Fable 5.1	Source	0.5859	0.6310	−4.51
	Source+MCP	0.6162	0.6344	−1.81
Claude Opus 5.5	Source	0.6569	0.6942	−3.73
	Source+MCP	0.6669	0.6881	−2.13

S4 contains 12 replicate record blocks (two models × two access conditions × three repeats), each with the same 94 IDs and one score per ID. S5 is restricted to those IDs from the 250-task FO export. Both report high effort, but identical question, reference, scorer, target and input builds have not been established. The subset contains 21 architecture, 14 biology, 17 chemistry, 4 CAD and 38 medical tasks. The GPT Codex control has 94 unique IDs and 94–179 attempt values per replicate block; its retry selection is unresolved. No matching GPT Claude Code high-effort control was located.

6.3.2 Fixed-view budget and camera plan

More fixed views do not produce a monotonic gain. In the historical cohort, aligned views change little from three to six (0.1516 to 0.1529) and decrease at twelve; orbit views tie at six and twelve (Table 4). Six views are a practical reference point for this sweep, not a demonstrated optimum for the current release.

The sweep uses GPT-6 Astra with Claude Code + LiteLLM on 250 v11 tasks, at reported `max` effort and with the last attempt retained. We use this export because it contains the complete aligned/orbit comparison at $N \in \{1, 3, 6, 12\}$; the current release supplies only an $N = 6$ anchor. The historical cohort shares 205 retained IDs with final250, excluding the 32 replacement and 13 later-replacement tasks. Its question, gold, scorer and camera manifests are not locked to the current build. We therefore retain the within-cohort pattern without splicing its scores into the current leaderboard. Appendix F gives the full diagnostics, current Fixed6 anchors and medical image-count exception.

Table 4: More fixed views do not produce a monotonic gain in the historical v11 sweep. Mean task scores for GPT-6 Astra with Claude Code + LiteLLM; these are historical design probes, separate from the current release.

Camera plan	$N = 1$	$N = 3$	$N = 6$	$N = 12$
Aligned	0.1223	0.1516	0.1529	0.1427
Orbit	0.1171	0.1204	0.1463	0.1463

Each cell averages the retained exported score over 250 v11 tasks, at reported max effort with the last attempt retained. N denotes the nominal view budget; historical medical inputs contain $2N$ image files. The current release provides only an $N = 6$ anchor, not a version-matched sweep. Full diagnostics and cohort limits appear in Appendix F.

6.3.3 Tool-only access versus source computation

Restricting tools asks a different question from adding MCP. Source+MCP permits both source computation and tool-based inspection; it evaluates that combined workflow. A declared MCP-only condition instead asks what the solver can do through the inspection interface. We retain the ninth arm of the historical v11 export because it records this distinct access protocol, which the current Source+MCP condition does not instantiate.

The historical GPT-6 Astra run uses Claude Code + LiteLLM, reported `max` effort and last-attempt retention, with 250/250 exported scores and a mean of 0.1757 (Table 5). The README declares an MCP-only tool surface, but tool lists, disabled file/shell-tool parameters, environment records and trajectories are unavailable. The current Source and Source+MCP rows provide access-protocol context, not a same-version pair with MCP-only. These records clarify the distinction between tool-restricted and combined access; they do not quantify the marginal benefit of MCP on the current release. Appendix F records the remaining isolation and provenance limits.

Table 5: MCP-only and Source+MCP represent different evidence-access protocols. Historical MCP-only record and current Astra source-access results. The route definitions, cohorts, harnesses, scorers and repeat policies differ; no same-version MCP-only pair is available.

Setting	Declared access	Harness	Mean
v11 MCP-only	Raw asset + MCP; MCP-only tool surface	Claude Code + LiteLLM	0.1757
Current Source	Source data + code execution	Codex	0.5291
Current Source+MCP	Source data + code execution + MCP	Codex	0.5282

All rows use GPT-6 Astra at reported max effort. **v11 (S1):** the ninth exported column has 250/250 scores and mean 0.1756667, retaining the last attempt. The README declares an MCP-only solver tool surface over the raw asset. The local package contains no tool list, disabled Bash/Read/Write parameters, environment record or trajectories, so file and tool isolation remain unverified. **Current (S2):** final250, score release `59dae5a6`; each task averages three scores. The `ccmcp` route provides source data, code and MCP and does not implement the historical MCP-only definition. The historical 0.1757 cannot be paired with the current 0.5282 to estimate an MCP effect.

6.4 Which scientific task requirements remain difficult?

Domain means reveal changes that cancel in the aggregate (Figure 4). For Astra, biology changes from 0.6200 to 0.6778 with MCP, while medical imaging changes from 0.3783 to 0.3406. Sol's biology score changes from 0.6000 to 0.6667. Among the canonical configurations in Figure 4, the highest domain mean is Fable 5.1's biology score with MCP, 0.6989. Appendix E provides all three condition-specific domain matrices.

Across the 36 canonical leaderboard configurations, 34 tasks have a mean score of zero everywhere. At least one configuration reaches a task mean of one on 133 tasks. The remaining 83 tasks have a best mean strictly between zero and one. Medical imaging contains 13 of the all-zero tasks, compared with four to six in each other domain (Figure 5). These profiles describe the tested configurations and questions; they

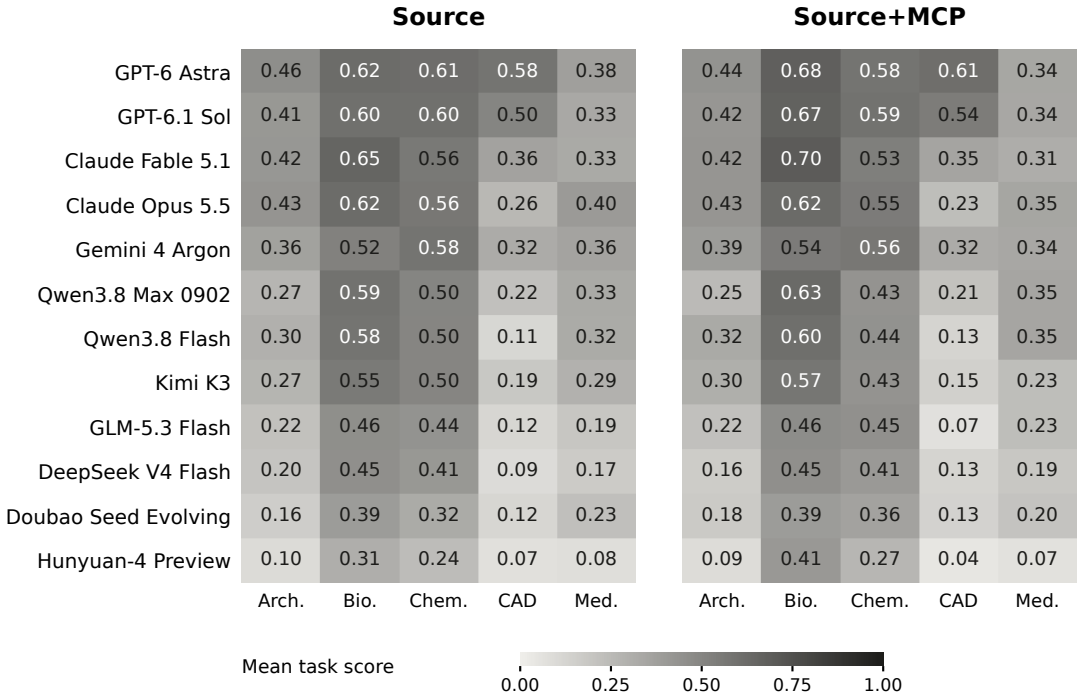


Figure 4: Domain mean task scores for the canonical Source and Source+MCP configurations. Each cell averages 50 tasks; darker cells indicate higher scores on a shared zero-to-one scale. Labels display two decimals. Tables 9–10 retain four decimals, and Table 11 provides Fixed6 domain results.

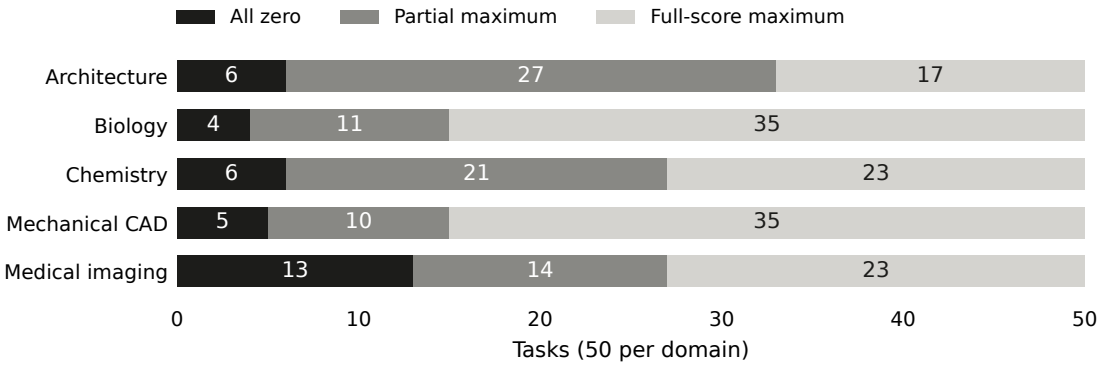


Figure 5: Task score profiles within each domain. Every task is assigned by its highest available-repeat mean across the 36 canonical configurations: zero (dark), strictly between zero and one (mid-grey), or one (light). Segment labels count tasks, with 50 tasks per row. These are descriptive categories for the evaluated cohort. Table 19 gives the counts.

do not establish intrinsic domain difficulty or diagnose observation and reasoning separately.

6.5 What can explain a failed answer?

Reference inspections demonstrate an intended route, rather than the model's actual observations. Historical pooled tool-use statistics remain separate from the current 36-configuration results and provide no human performance baseline.

6.6 How sensitive are the findings to reference quality?

A prior targeted audit of eight CAD questions identified underspecified targets or counting units without establishing a numerical gold error. We retain the published 250-task means and separately recompute scores after excluding those questions. The resulting 242-task diagnostic preserves all three complete model rankings, every Source–Fixed6 direction, and the eight-increase/four-decrease MCP split.

Astra's Source score changes from 0.5291 to 0.5356; Fable's Fixed6 score changes from 0.2965 to 0.3063. Appendix E lists the excluded IDs and complete comparison. This sensitivity check neither replaces the benchmark nor validates all remaining references.

7 Discussion and Limitations

ScienceVision evaluates specified judgments about supplied digital artifacts. It does not measure the ability to formulate research questions, design experiments, or validate a new scientific claim. A task can require scientific knowledge without representing the full work of a scientist. The title's broad question must be interpreted through this operational scope.

Equal domain quotas make the aggregate contributions equal, but do not represent the frequency of questions in scientific practice. Subdomains and asset families remain uneven. Task selection has used prior model performance, so measured difficulty is conditional on the selection procedure and not an estimate for randomly sampled scientific questions. A human performance baseline has not been established by the reference inspections or annotation hours.

Reference provenance and accepted-answer rules affect the meaning of a score. Recorded values, script-derived references, and tool-derived references provide different forms of evidence, and none guarantees correctness solely by being present in the release. Model–harness differences, mixed run segments, incomplete repeats, and uncertain historical enforcement further limit attribution of measured differences to a single capability.

The current evaluation grades answers. The released task-level scores do not reveal which components passed unless component verdicts are retained. They also cannot by themselves separate evidence-acquisition failure from incorrect domain inference. That distinction requires trajectory-linked observations and validated process annotations. The Source+MCP condition provides an interface for such work; its availability and aggregate score are not proof that it occurred.

8 Conclusion

ScienceVision organizes scientific questions about native 3D artifacts around explicit answer contracts and task-specific references across five domains. The three evidence-access conditions distinguish source computation, source access with inspection tools, and fixed rendered inputs. Current scores characterize evaluated model–harness configurations; they do not independently establish successful observation, correct reasoning processes, or a causal benefit from MCP.

Contributors

Core contributors: Tianyi Ma[1,*], Jing Gu[2,*], Haozhe Zhao[1,‡], Liang Chen[1,‡], Yunfan Zhang[1], Haijun Su[2], Jialong Wu[1], Le Xue[2], Arnav Das[2], Dustin Tran[2], Yinfei Yang[2], Andrew Dai[2,†], Kuan Li[1,†].

Contributors: Liangkun He[1], Zihao Guo[1], Xiang Gong[1], Yutao Zhou[1], Jiarui Sun[1], Chengda Wu[1], Zhao Zhang[1], Qianhui Yang[1], Zhenji Kang[1], Yi Shen[1], Wenda Wang[2].

Affiliations: [1]UniPat AI. [2]Elorian AI.

[*]Equal contribution. [‡]Project leader. [†]Corresponding author.

References

Kuangshi Ai, Haichao Miao, Kaiyuan Tang, Nathaniel Gorski, Jianxin Sun, Guoxi Liu, Helgi I. Ingolfsson, David Lenz, Hanqi Guo, Hongfeng Yu, Teja Leburu, Michael Molash, Bei Wang, Tom Peterka, Chaoli Wang, and Shusen Liu. SciVisAgentBench: A benchmark for evaluating scientific data analysis and visualization agents, 2026. URL https://arxiv.org/abs/2603.29139v4.

Anthropic. Claude code documentation, 2026. URL https://code.claude.com/docs/en/overview.

Yixiong Chen, Wenjie Xiao, Pedro R. A. S. Bassi, Boyan Wang, Liang He, Xinze Zhou, Sezgin Er, Ibrahim Ethem Hamamci, Zongwei Zhou, and Alan Yuille. DeepTumorVQA: A hierarchical 3d CT benchmark for stage-wise evaluation of medical VLMs and tool-augmented agents. *arXiv preprint arXiv:2605.09679*, 2026.

egtai. GMX-VMD-MCP: MCP service for GROMACS and VMD molecular dynamics simulations and visualization. Software repository, 2025. URL https://github.com/egtai/gmx-vmd-mcp. Accessed 2026-09-23.

Lawrence Livermore National Laboratory. BioImage-Agent. Software repository, 2025. URL https://github.com/LLNL/bioimage-agent. Accessed 2026-09-23.

Shusen Liu, Haichao Miao, Zhimin Li, Matthew Olson, Valerio Pascucci, and Peer-Timo Bremer. AVA: Towards autonomous visualization agents through visual perception-driven decision-making. *Computer Graphics Forum*, 43(3):e15093, 2024. doi: 10.1111/cgf.15093. URL https://doi.org/10.1111/cgf.15093.

Shusen Liu, Haichao Miao, and Peer-Timo Bremer. ParaView-MCP: An autonomous visualization agent with direct tool use, 2025. URL https://arxiv.org/abs/2505.07064.

Aravind Mandiga, Guoming Li, Jin Lu, Ismailcem Budak Arpinar, Khaled Rasheed, and Samuel E. Aggrey. ProtStructQA: A denotation threshold in protein structural reasoning, 2026. URL https://arxiv.org/abs/2606.00451.

Giorgio Mariani, Alessandro Raganato, Simone Melzi, and Gabriella Pasi. GLUE3D: General language understanding evaluation for 3d point clouds. *Information Fusion*, 129:104007, 2026. doi: 10.1016/j.inffus.2025.104007.

OpenAI. Codex CLI, 2026. URL https://developers.openai.com/codex/cli.

Tom Peterka, Tanwi Mallick, Orcun Yildiz, David Lenz, Cory Quammen, and Berk Geveci. ChatVis: Large language model agent for generating scientific visualizations, 2025. URL https://arxiv.org/abs/2507.23096.

Mingjia Wang, Taiting Lu, Ziwei Dong, Sisong Bei, Jingying Zeng, Runze Liu, Kaiyuan Lin, Hongxing Pan, Kai Zhang, Yizheng Hou, Yangshoudu Zheng, Chenchen Guo, Weiyuan Meng, Shubin Lyu, Zhijun Zheng, Dexu Wang, Xinyu Bai, Shurui Qian, Zhangzixin, Mengyu Pan, Guoliang Shi, Ling Ma, Yifan Yang, Qi He, Yi-Chao Chen, Yincheng Jin, Sung-Liang Chen, and Mahanth Gowda. OmniCAD: A large-scale benchmark for 3D spatial reasoning in robotics assemblies, 2026. URL https://arxiv.org/abs/2608.22637.

zhaoyouj. MCP-Slicer: 3D Slicer model context protocol integration. Software repository, 2025. URL https://github.com/zhaoyouj/mcp-slicer. Accessed 2026-09-23.

A Task Specifications and Coverage

A.1 Subdomain inventory

Table 6 gives the 22 subdomains and their task counts. These are the descriptive groups attached to the selected dataset. Each task has one listed subdomain; a task can involve several observations or judgments. For example, a conformation task can also ask for an occluded atom. The groups are not mutually exclusive definitions of scientific abilities. The accompanying metadata records the task-to-subdomain mapping and question hashes.

Table 6: Complete subdomain composition. Objectives are illustrative descriptions of question content; they are not additional per-task ability annotations.

Subdomain	Tasks	Illustrative objective
Architecture (50)		
Dimensional compliance	16	Check dimensions against a stated requirement.
Alignment and stacking	12	Identify continuity or alignment of members.
Interference and clearance	12	Distinguish spatial interference from clearance.
Object geometry readings	10	Read object dimensions, arrangement, or scene geometry.
Biology (50)		
Molecular contact	14	Identify atom contacts and associated distances.
Spatial arrangement	14	Describe the relative arrangement of components.
Interaction typing	8	Identify interaction classes or coordinating atoms.
Proximity choice	8	Select the closest specified component or atom.
Occluded target geometry	6	Locate a view-defined target and assess its geometry.
Chemistry (50)		
Conformation and occlusion	21	Combine molecular conformation with occluded targets.
Dihedral measurement	9	Measure a specified torsion or its minimum value.
Molecular extent	9	Determine the separation of farthest heavy atoms.
Hybridization state	7	Classify and count heavy-atom hybridization states.
Crystal periodicity	4	Analyze a unit cell and its periodic contents.
Mechanical CAD (50)		
Part-instance counting	20	Count component copies, including hidden instances.
Part-property identification	14	Determine a part's geometric properties.
Assembly-composition counting	10	Count repeated parts or designs within an assembly.
Mechanism motion	6	Analyze gears, mechanisms, or assembly relations.
Medical imaging (50)		
Lesion pointing	25	Locate a specified abnormality in a CT volume.
Slice comparison	12	Compare visible anatomy across specified slices.
Geometry measurement	8	Measure a structure's extent or position.
Anatomy pointing	5	Locate a named anatomical structure in a volume.
Total	**250**	**22 subdomains**

A.2 Answer structure and conventions

The normalized references contain 143 answers with one top-level field and 107 with multiple fields. The counts for 1, 2, 3, 4, 5, 6, 7, and 9 fields are 143, 35, 45, 11, 5, 8, 2, and 1, respectively. Normalization can wrap a single-word response in an `answer` field. A list-valued field remains one field. These counts do not segment a question into independently requested outcomes.

Contracts can specify an atom identity, an ordered list, a categorical answer, or a coordinate or measurement. The original question determines the convention: voxel extents need not have the same unit as physical distances, and a list of intermediate atoms need not contain its endpoints. The following examples preserve the stored reference values. They illustrate the intended answers, without defining numerical tolerances or equivalent accepted answers.

A.3 Reference inspection operations

All 250 tasks have a recorded expert reference inspection. We count records in `trajectory.steps` that contain a tool operation and exclude the initial `step:0, tool:null` state. The current records contain 2,156 operations. The ranges 1–2, 3–5, 6–10, 11–20, 21–40, and 41 or more operations contain 36, 95, 62, 32, 21, and 4 tasks, respectively. The accompanying task metadata preserves each count and the corresponding trajectory SHA-256. These counts describe recorded viewer operations, not model runtime calls or a minimum reasoning length.

A.4 Chemistry example: complete question and reference

Task `sv-chem-chem_conformation-0002`. The question text, preceding its answer-format block, is:

> A molecular conformation is displayed in the viewer. Inspect all dihedral angles formed by four consecutively bonded atoms with the element sequence H-C-O-C. Identify the dihedral angle with the smallest absolute value and report its absolute value in degrees.
>
> From the initial camera view, identify the atom that is almost completely occluded and the heavy atom that is partially occluded. Determine the shortest bonded path between these two atoms and report the ordered atom IDs of the atoms strictly between them, excluding the two endpoint atoms themselves, together with the number of those intermediate atoms. If the shortest path contains at least four atoms in total, examine every consecutive four-atom segment along that path and report the smallest absolute dihedral angle. If the path contains fewer than four atoms, report null for the path dihedral.

The requested JSON fields and their stored reference values are:

```
{
  "minimum_absolute_dihedral_deg": 53.34,
  "nearly_fully_occluded_atom_id": 21,
  "partially_occluded_heavy_atom_id": 2,
  "shortest_bond_path_atom_ids": [8, 9, 3],
  "shortest_bond_path_internal_atom_count": 3,
  "shortest_path_minimum_absolute_dihedral_deg": 64.49
}
```

The two angle fields answer different requests: the first ranges over H-C-O-C segments in the molecule, and the last ranges over consecutive four-atom segments on the selected path. Figure 2, panel (a), shows the initial view; panel (b) shows the later reference inspection with labels and measurements.

A.5 Medical example: complete question and reference

Task `sv-medi-med_ct_geometry-0103`. The question text, preceding its answer-format block, is:

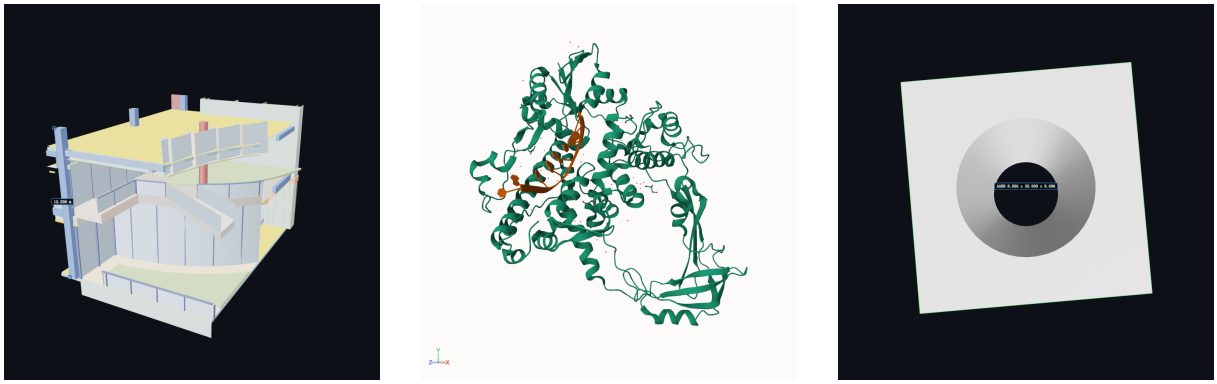
> The target is the small amount of left pleural effusion visible in this CT volume. A bounding box extent includes both bounding voxels, so along each axis it equals the difference between the two bounds plus 1. The sampling grid's three axes are numbered 0, 1 and 2. If several axes tie for widest, report all of them in increasing order rather than picking one. No measurement has more than 6 decimal places. Find a target's centroid and its representative slice. Scroll through the selected overlay and place the centroid. Measure target extent on all three array axes, identify every longest array axis and measure signed left-right centroid offset.

The requested JSON fields and their stored reference values are:

```
{
  "extent_vox": [86, 81, 61],
  "longest_axes": [0],
  "signed_lr_distance_mm": -64.56
}
```

The maximum extent occurs on array axis 0. The signed offset is reproduced as recorded; the axis-index list and voxel extents should not be relabeled as anatomical directions or physical lengths without the corresponding coordinate definition.

A.6 Examples from architecture, biology, and mechanical CAD



(a) Architecture. (b) Biology. (c) Mechanical CAD.

Figure 6: Current-dataset examples from the remaining three domains. The original reference inspection images are frame 1 for architecture, frame 1 for biology, and frame 3 for mechanical CAD. These selected states illustrate the supplied objects.

Architecture: continuity across storeys. Task `arch-anzj-align-a07` asks which vertical structural members extend continuously across multiple storeys, how many there are, their shortest and longest vertical extents, and whether discontinuous members exist. A continuous member must geometrically span at least two storeys without interruption over that range. The requested coordinates are in millimetres with Z up. The question supplies the conversion from the viewer's metre-scale, Y-up glTF frame:

$$\text{rhino}_{\text{mm}} = (x_{\text{glTF}}, -z_{\text{glTF}}, y_{\text{glTF}}) \times 1000.$$

The reference contains `continuous_member_count:"4"`, `shortest_continuous_extent_mm:"12000"`, `longest_continuous_extent_mm:"12200"`, and `discontinuous_member_exists:"false"`.

Biology: terminal residues and chain position. Task `sv-biol-bio_collision_check-0015` asks whether the N-terminus or C-terminus of a specified polymer chain is closer to its centre. The subject is the chain with the most atoms, with ties resolved by the earliest chain identifier. The question represents the centre by the alpha carbon of the middle residue, taking the later middle residue for an even-length chain. It represents the termini by the alpha carbons of the first and last residues in file order. The task uses the first model, retains the highest-occupancy alternate positions, and excludes hydrogens and solvent. The reference field `terminus_nearer_chain_centre` stores `"C-terminus"`, one of the two labels requested by the question.

Mechanical CAD: a component's geometry. Task `mv-073` asks whether a displayed component is essentially flat or bends substantially out of any single plane. It requests exactly one word, `planar` or `non-planar`. The normalized reference stores `"non-planar"` under `answer`. This example shows that a single-field normalized record can correspond to an unstructured one-word response.

B Construction, References, and Scoring

B.1 Task-specific scoring and verification scope

Equation 1 defines the scoring interface. The available task configurations specify parsing, aliases, nested objects, comparison rules, component weights, and explicit overrides. Some overrides score only a subset of stored answer fields. The configuration inventory does not include the complete external default comparator implementation, so it cannot establish a universal field average or eligibility gate. Our aggregation retains the exported scores without independently applying such a gate.

C Access Interfaces and Fixed Views

The interface supports camera control, selection and isolation, clipping or slice navigation, and scientific measurements, with available operations depending on the viewer. These operation classes describe the interface design. Expert-browser operations and agent-side tools can have different names or only partial counterparts. Released allow lists identify permitted tool names; they do not verify run-specific exposure or successful server operation. Source+MCP also permits computation from source data, so its score does not identify whether a viewer supplied the decisive evidence.

D Configurations, Versions, and Repeat Coverage

The main result table, domain matrices, paired contrasts, score profiles, and CAD exclusion diagnostic are generated from a repository-contained frozen export. It contains 9,000 task–configuration records: 250 shared task IDs for each of 12 models under three conditions. The task release is `bb27be96c10c8629ce2395b55a98b80e8bfb9c72`; the score release is `59dae5a61aea98bc225a6a306a8b329762ea1b69`.

The score CSV SHA-256 is

`d104e3cd96ddc94bf36a36c4b17f1fb7ff27239f6e4eb7239d9590227cda32e0`.

The local manifest records checksums for the compact inputs and their upstream provenance. The generator checks unique task membership, domain labels, available-repeat counts, and the equivalence of the stored task mean to the mean of exported repeats. It reproduces all 36 four-decimal released aggregates. No unavailable repeat is treated as an extra zero, and no intermediate rounding is applied.

The local blog snapshots reproduce the same 36 displayed main scores. They round each task mean to four decimals before computing domain means. Preserving full task-mean precision changes one displayed four-decimal domain cell: Astra Source/chemistry is 0.6094 here, versus 0.6095 under the blog rule. This is a rounding difference, not a new evaluation. These checks reproduce exported scores rather than replaying the agent runs or independently verifying runtime scorer behavior.

D.1 Configuration declarations and incomplete run metadata

The study setup assigns Codex to GPT-6 Astra and GPT-6.1 Sol and Claude Code to the other ten models across all three conditions. This declaration agrees with the selected Source and Source+MCP rows. GPT Codex uses requested reasoning effort `max`. Canonical Claude Code effort is not established by the score exports. The rerun declaration also assigns the leaderboard harnesses to the 39-task rerun across all three conditions. The Fixed6 score summary has no per-row harness field, so the declarations do not independently reconstruct every retained image run's execution system.

Release documentation describes retaining earlier outputs and rerunning 39 tasks after question or scoring revisions. This rerun is distinct from task-membership provenance: the frozen score records contain 205 retained-v11 IDs, 32 replacement IDs, and 13 later-replacement IDs per configuration. Task-set intersections alone do not certify unchanged questions, references, or run inputs. Model aliases also do not uniformly identify fixed model versions. Release-level consistency checks cannot reconstruct each retained run's scorer and input build.

The three-repeat protocol yields 26,653 available exported repeat scores. The exceptions are Gemini under both source-access conditions (172 tasks with two repeats and 78 with three) and Hunyuan Fixed6 (three tasks with two repeats and 247 with three). All other cells have three scores per task. These exports verify score availability, not independence between repeats.

The supplementary Fable/Opus Codex batch requests `high` effort. The authors report choosing it because `max` API calls encountered timeouts. This model-specific setting is not an equalized reasoning budget.

Completion-token counts and interaction steps are reported separately from reasoning tokens, which are unavailable for this batch.

D.2 Cross-harness source layers

Tables 2 and 3 use the following exports. S2 is the current `score-records.csv` identified above. S3 is the 2026-10-17 consolidated export, using the `r1_claudecode`, `r2_claudecode` and `r3_claudecode` columns of the Astra/Sol `cc` and `ccmcp` per-task files. S5 is `fo-codex-fable-opus-20261020.csv`, containing 3,000 score rows: two models × two access conditions × three repeats × 250 tasks. Its frozen normalized copy removes only the internal score-file paths; the manifest preserves both original and normalized checksums. All 16 configuration cells have the same 250 IDs and three scores per task, but matching IDs does not establish matching question, gold, scorer or input builds.

S4 is `ctl94-claude-fable-opus-20261020.json`; its 12 replicate record blocks each contain the exact `fix94.txt` ID set and one score per ID. Restricting S5 to that set produces the four Codex comparison cells. S4's high-effort label comes from the accompanying comparison-script declaration. S5's requested `high` setting is independently recorded in all 3,000 trial configurations. The subset was selected for the fix round: 21 architecture, 14 biology, 17 chemistry, 4 CAD and 38 medical tasks. Its means weight tasks equally, not domains equally. Their original question, target, reference, scorer and input builds have not been matched to each other or to the final release.

S6, `ctl94-codex-gpt-20261020.json`, contains 94 unique IDs per replicate block but 94–179 stored attempt values. A retry-selection rule and a matching GPT Claude Code high-effort control are unavailable. A candidate rescore overlay, `rescore_overlay_20261021.json`, records the blog's GPT Claude Code aggregates 0.4402/0.4262 and 0.4185/0.4163, but those means are not reproducible from the available trial-level corrections. They are excluded from the checked tables, which retain S3 and calculate percentage-point differences before rounding. Necessary score exports and source declarations are frozen in the repository. Their manifest records original and archived SHA-256 values and the explicit S5 path-column removal. The unified generator checks all 16 full-set and eight selected-subset cells, without replaying runs or applying a new eligibility gate.

D.3 Interaction length, tool feedback, and timeout reliability

The archived Fable/Opus Codex FO batch provides 3,000 trajectories and matching trial results for the S5 score export. All 3,000 trial configurations record a requested effort of `high`. All 3,000 have parseable trajectories and completion-token telemetry. We count agent response entries as trajectory entries whose `source` is `agent`, including the final answer and excluding system/user initialization. Tool-call counts sum the calls in those entries. Completion-token counts use `final_metrics.total_completion_tokens`; the reported Claude `reasoning_output_tokens` field is zero throughout this batch and does not provide a measurement of hidden reasoning length.

The 1,243 full-score trials have a median of six agent response entries and 2,419 completion tokens; the 1,061 zero-score trials have medians of nine agent response entries and 4,628 tokens. The other 696 trials receive partial scores. These are descriptive score-group associations across tasks, models and access conditions, not a test that shorter traces are better. There are 47 `AgentTimeoutError` records out of 3,000 trials, each recording a 3,600-second execution limit. These executor timeouts are distinct from the API timeout behavior that motivated the authors' `high` setting for Claude Codex. The older attribution report also describes a `high-to-max` pilot, but its original pairing and denominator were not reconstructed here; its precise rates are not used in the manuscript.

One recovery example is Fable/Codex Source repeat 3 on `sv-chem-chem_binding_geometry-0114`. Recorded trajectory entry 4 fails to import `gemmi`. Entries 5 and 6 inspect the CIF atom-table columns and entity metadata; entry 7 builds a replacement parser and selects the largest occupied, non-water,

non-polymer component. Entry 8 computes the closest atom-pair distance, and entry 9 submits 2.229 angstroms, which receives full credit under the archived rubric. The trace contains six agent entries (recorded IDs 4–9, after initialization) and five tool calls. It shows recovery through additional tool feedback and computation, without establishing a causal effect of trace length or a difference from Claude Code. The selected Mg–histidine coordination pair is not excluded because the CIF has no `struct_conn` records. This reproduces the archived calculation and score, rather than certifying the semantic definition of a nonbonded contact.

The qualitative interface comparison uses the shell-execution calls in the FO Codex records and Bash, Read and occasional Edit calls in the available Claude Code control records. It concerns tool granularity and orchestration, not an independently measured benefit from call ordering. The canonical Claude Code score-only records do not provide matching trajectories for that attribution. The older `harness-attribution.zh.html` provides the analysis questions, but its claims about absent effort controls, older leaderboard and scorer-repair decomposition are not transferred to the current comparison.

E Complete Results and Sensitivity Analyses

E.1 Canonical scores and execution declarations

Table 7: Canonical mean task scores for 12 models under three access settings on the same 250 tasks. Available repeats are averaged within each task, then tasks are weighted equally. Values are displayed directly to four decimals. Complete setting definitions and execution declarations follow below; supplementary cross-harness runs are separate from these selected leaderboard rows.

Model	Source	Source+MCP	Fixed6
GPT-6 Astra	0.5291	0.5282	0.2498
GPT-6.1 Sol	0.4893	0.5110	0.2044
Claude Fable 5.1	0.4619	0.4631	0.2965
Claude Opus 5.5	0.4539	0.4343	0.2383
Gemini 4 Argon	0.4278	0.4321	0.1462
Qwen3.8 Max 0902	0.3830	0.3736	0.2450
Qwen3.8 Flash	0.3616	0.3681	0.2468
Kimi K3	0.3609	0.3369	0.2148
GLM-5.3 Flash	0.2846	0.2858	0.1521
DeepSeek V4 Flash	0.2652	0.2678	0.1661
Doubao Seed Evolving	0.2435	0.2527	0.1102
Hunyuan-4 Preview	0.1598	0.1738	0.0824

Access settings. **Source** provides native source files and code execution. **Source+MCP** adds MCP inspection and measurement tools to the same source-file and code access; their use is permitted, not required. **Fixed6** provides six fixed rendered view states instead of native source-file access. The designated medical package contains paired slice/render images (12 files for six view states); this package definition is not an audit of every retained run's actual input.

Execution declarations. The study assigns Codex to both GPT models and Claude Code to the other ten across all three settings. Table 8 records the exact model identifiers and known source-access effort. Fixed6 per-row runtime harness and effort are absent from its score export; the study-level declaration is not a row-level execution audit. Canonical Claude Code effort is not recorded. Repeat exceptions and comparison limits are in Section 5 and Appendix D.

Table 8: Recorded model identifiers, declared harness for all three access settings, and source-access reasoning effort. Row order matches Table 7; the declarations do not establish fixed model versions, complete execution budgets or retained-run provenance.

Recorded model identifier	Declared harness	Source-access effort
`gpt-6-astra`	Codex	max (reported)
`gpt-6.1-sol`	Codex	max (reported)
`claude-fable-5-1`	Claude Code	Not recorded
`claude-opus-5-5`	Claude Code	Not recorded
`gemini-4-argon`	Claude Code	Not recorded
`qwen3.8-max-0902`	Claude Code	Not recorded
`qwen3.8-flash`	Claude Code	Not recorded
`kimi-k3`	Claude Code	Not recorded
`glm-5.3-flash`	Claude Code	Not recorded
`deepseek-v4-flash`	Claude Code	Not recorded
`doubao-seed-evolving`	Claude Code	Not recorded
`hy4-preview`	Claude Code	Not recorded

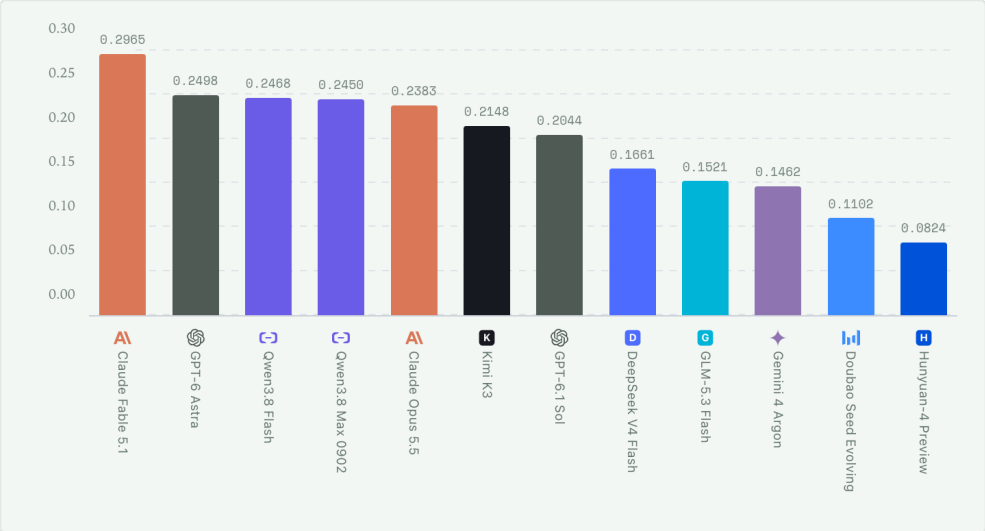


Figure 7: Fixed6 leaderboard for the same 12 canonical configurations: six fixed rendered view states. This figure is re-exported from the local blog source using the unrounded task means in Table 7; full model names and scores to four decimals are shown directly. Source and Source+MCP are shown in the main text (Figure 3).

E.2 Condition-specific domain scores

Tables 9–11 report separate domain matrices for the current three conditions. Each cell averages 50 tasks after averaging available repeats within each task. Because the five domains have equal size, their unrounded means reconstruct the 250-task mean. The columns describe the selected questions and configurations rather than population-level estimates of disciplinary difficulty. Historical subdomain scores averaged over two routes are excluded from these matrices.

Table 9: Source domain mean task scores, with 50 equally weighted tasks per domain. Model configurations follow Table 7. Values retain full task-mean precision until display; some last digits differ from the blog, which rounds task means before aggregation.

Model	Arch.	Biology	Chemistry	CAD	Medical
GPT-6 Astra	0.4576	0.6200	0.6094	0.5800	0.3783
GPT-6.1 Sol	0.4132	0.6000	0.6006	0.5000	0.3328
Claude Fable 5.1	0.4161	0.6500	0.5572	0.3600	0.3261
Claude Opus 5.5	0.4283	0.6178	0.5639	0.2600	0.3994
Gemini 4 Argon	0.3638	0.5167	0.5811	0.3200	0.3572
Qwen3.8 Max 0902	0.2715	0.5933	0.5028	0.2200	0.3272
Qwen3.8 Flash	0.2959	0.5789	0.4995	0.1133	0.3206
Kimi K3	0.2744	0.5511	0.4995	0.1867	0.2928
GLM-5.3 Flash	0.2178	0.4556	0.4361	0.1200	0.1933
DeepSeek V4 Flash	0.2008	0.4511	0.4106	0.0933	0.1700
Doubao Seed Evolving	0.1582	0.3922	0.3222	0.1200	0.2250
Hunyuan-4 Preview	0.0955	0.3067	0.2417	0.0733	0.0817

Table 10: Source+MCP domain mean task scores, with 50 equally weighted tasks per domain. Model configurations follow Table 7. Values retain full task-mean precision until display; some last digits differ from the blog, which rounds task means before aggregation.

Model	Arch.	Biology	Chemistry	CAD	Medical
GPT-6 Astra	0.4356	0.6778	0.5806	0.6067	0.3406
GPT-6.1 Sol	0.4211	0.6667	0.5911	0.5400	0.3361
Claude Fable 5.1	0.4243	0.6989	0.5295	0.3533	0.3094
Claude Opus 5.5	0.4250	0.6211	0.5467	0.2333	0.3456
Gemini 4 Argon	0.3942	0.5450	0.5639	0.3200	0.3372
Qwen3.8 Max 0902	0.2480	0.6289	0.4300	0.2133	0.3478
Qwen3.8 Flash	0.3242	0.6022	0.4389	0.1267	0.3483
Kimi K3	0.3023	0.5722	0.4311	0.1467	0.2322
GLM-5.3 Flash	0.2217	0.4611	0.4489	0.0667	0.2306
DeepSeek V4 Flash	0.1647	0.4522	0.4061	0.1267	0.1894
Doubao Seed Evolving	0.1751	0.3911	0.3606	0.1333	0.2033
Hunyuan-4 Preview	0.0870	0.4078	0.2683	0.0400	0.0661

Table 11: Fixed6 domain mean task scores, with 50 equally weighted tasks per domain. Model configurations follow Table 7. Values retain full task-mean precision until display; some last digits differ from the blog, which rounds task means before aggregation.

Model	Arch.	Biology	Chemistry	CAD	Medical
GPT-6 Astra	0.1500	0.3000	0.1756	0.3400	0.2833
GPT-6.1 Sol	0.1385	0.2567	0.1400	0.2667	0.2200
Claude Fable 5.1	0.2387	0.4911	0.3022	0.1800	0.2706
Claude Opus 5.5	0.2269	0.2978	0.2578	0.2133	0.1956
Gemini 4 Argon	0.1008	0.2844	0.0889	0.1400	0.1167
Qwen3.8 Max 0902	0.1885	0.4600	0.1894	0.2067	0.1806
Qwen3.8 Flash	0.1934	0.5467	0.1705	0.1533	0.1700
Kimi K3	0.1678	0.4611	0.1400	0.1600	0.1450
GLM-5.3 Flash	0.1212	0.3533	0.0978	0.1067	0.0817
DeepSeek V4 Flash	0.1129	0.3367	0.1067	0.1533	0.1211
Doubao Seed Evolving	0.0489	0.2433	0.0689	0.1267	0.0633
Hunyuan-4 Preview	0.0100	0.3333	0.0022	0.0467	0.0200

E.3 Subdomain Results and Aggregation

For these descriptive cross-model comparisons, we average the 12 canonical models equally within each access condition.

Source–Fixed6 gaps depend on the task family. The two largest cross-model gaps occur in dihedral measurement ($N = 9$; 45.06 percentage points) and conformation/occlusion ($N = 21$; 40.68 points). Every model scores higher with Source in both subdomains. Biological spatial arrangement ($N = 14$) has a much smaller mean gap of 2.48 points, and 5 models score higher with Fixed6.

The Source+MCP difference changes sign within every model. Each of the 12 models has both positive and negative Source+MCP minus Source differences across the 22 subdomains. Molecular contact ($N = 14$) has a mean difference of +8.04 percentage points, with 9 models higher. Dihedral measurement ($N = 9$) has a mean difference of -5.25 points, with 8 models lower. A single overall MCP difference conceals these opposing local changes.

Lesion pointing remains a shared weakness. On the 25 lesion-pointing tasks, mean scores are 0.1011, 0.0967 and 0.0344 for Source, Source+MCP and Fixed6, respectively. All 36 model–condition configurations score lower on lesion pointing than on slice comparison ($N = 12$). The medical domain mean mixes this localization weakness with higher slice-comparison performance.

Data and aggregation. The six tables report 792 means over 22 subdomains for 12 models and three conditions. Source, Source+MCP and Fixed6 are kept separate. Scores use the frozen score release `59dae5a6`, including the retained and 39-task rerun segments used in the current configuration table. Subdomain assignments come from the corresponding 250-task release `bb27be96`. Each task has one native `sub_domain` category, preserved in the repository's dataset profile and frozen task-to-subdomain mapping. These categories describe task content, not mutually exclusive reasoning abilities; they are distinct from the release's separate `direction` labels.

For task i, we first average its available exported repeat scores, $\bar{s}_i = R_i^{-1} \sum_{r=1}^{R_i} s_{ir}$. A subdomain u with N_u tasks has mean $S_u = N_u^{-1} \sum_{i \in u} \bar{s}_i$. The domain and overall means are reconstructed as

$$S_d = \sum_{u \subset d} \frac{N_u}{50} S_u, \qquad S = \sum_u \frac{N_u}{250} S_u.$$

The five domains each contain 50 tasks, so S also equals the arithmetic mean of the five domain scores. The 22 subdomains contain 4 to 25 tasks each; averaging their means without task-count weights would change the statistic. All computations retain full precision until the four-decimal display.

Every model–condition pair covers all 250 tasks, and all 792 subdomain cells are populated. Gemini has two available repeats for 172 tasks and three for 78 tasks under each source-access condition. Hunyuan Fixed6 has two repeats for three tasks and three for 247; all other task–configuration records have three. Missing repeats are not replaced with zero, and repeats are not pooled across tasks. Exported zero scores remain in the denominator. The tables retain each route separately, with no averaging of Source and Source+MCP. Their unrounded weighted totals reproduce all 180 domain means and 36 overall means in the current release.

The Source rows use Codex for Astra and Sol and Claude Code for the other models. Fixed6 per-row harness metadata is not established by its score summary. Other model-version, retained/rerun and scoring limitations remain those of the canonical release. Access-condition differences compare evaluated configurations and do not isolate causal effects of MCP or active inspection. Small subdomain counts describe this selected benchmark and do not establish population-level differences in disciplinary difficulty.

Table 12: Source subdomain mean task scores, model group 1 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.2639	0.2593	0.2569	0.3125	0.2014	0.3056
Dimensional compliance	16	0.4071	0.3551	0.3208	0.3781	0.3521	0.1821
Interference and clearance	12	0.3361	0.3222	0.3685	0.3569	0.3451	0.1801
Object geometry readings	10	0.9167	0.8000	0.8167	0.7333	0.6000	0.4833
Domain mean	50	0.4576	0.4132	0.4161	0.4283	0.3638	0.2715
Biology							
Interaction typing	8	0.8750	0.8750	0.7500	0.7500	0.1250	0.7500
Molecular contact	14	0.7857	0.7143	0.8333	0.8333	0.8214	0.7381
Occluded target geometry	6	0.0000	0.0000	0.2222	0.0926	0.0000	0.2778
Proximity choice	8	0.5000	0.5000	0.5833	0.4583	0.6250	0.5000
Spatial arrangement	14	0.6429	0.6429	0.6310	0.6429	0.5952	0.5476
Domain mean	50	0.6200	0.6000	0.6500	0.6178	0.5167	0.5933
Chemistry							
Conformation occlusion	21	0.6601	0.6283	0.5701	0.5886	0.6905	0.5675
Crystal periodicity	4	0.7222	0.7778	0.7222	0.7222	0.7500	0.6111
Dihedral measurement	9	0.5556	0.5556	0.4630	0.5556	0.5556	0.5370
Hybridization state	7	0.8095	0.7778	0.9048	0.7778	0.5794	0.4921
Molecular extent	9	0.3395	0.3642	0.2778	0.2778	0.2778	0.2778
Domain mean	50	0.6094	0.6006	0.5572	0.5639	0.5811	0.5028
Mechanical CAD							
Assembly composition count	10	0.4333	0.4000	0.3000	0.2000	0.4833	0.1667
Mechanism motion	6	0.5000	0.5000	0.5000	0.4444	0.2778	0.3333
Part instance count	20	0.5667	0.3667	0.1667	0.0500	0.0750	0.0667
Part property identification	14	0.7381	0.7619	0.6190	0.5238	0.5714	0.4286
Domain mean	50	0.5800	0.5000	0.3600	0.2600	0.3200	0.2200
Medical imaging							
Anatomy pointing	5	0.8000	0.8000	0.4667	0.6667	0.6667	0.6667
Geometry measurement	8	0.5938	0.5799	0.4965	0.5382	0.5035	0.5243
Lesion pointing	25	0.2133	0.1467	0.1600	0.1867	0.1333	0.1467
Slice comparison	12	0.4028	0.3611	0.5000	0.6389	0.5972	0.4306
Domain mean	50	0.3783	0.3328	0.3261	0.3994	0.3572	0.3272
Overall	250	**0.5291**	**0.4893**	**0.4619**	**0.4539**	**0.4278**	**0.3830**

Table 13: Source subdomain mean task scores, model group 2 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.2569	0.2315	0.2116	0.1505	0.1829	0.1319
Dimensional compliance	16	0.2886	0.3179	0.2585	0.2177	0.1836	0.0275
Interference and clearance	12	0.3551	0.3074	0.2125	0.2014	0.1343	0.1319
Object geometry readings	10	0.2833	0.2167	0.1667	0.2333	0.1167	0.1167
Domain mean	50	0.2959	0.2744	0.2178	0.2008	0.1582	0.0955
Biology							
Interaction typing	8	0.7500	0.6250	0.1250	0.1250	0.0833	0.1250
Molecular contact	14	0.7619	0.7143	0.7619	0.7143	0.6905	0.5238
Occluded target geometry	6	0.1296	0.0370	0.1296	0.0926	0.0741	0.0000
Proximity choice	8	0.4583	0.5417	0.4583	0.4167	0.3333	0.3750
Spatial arrangement	14	0.5595	0.5714	0.4762	0.5476	0.4405	0.2857
Domain mean	50	0.5789	0.5511	0.4556	0.4511	0.3922	0.3067
Chemistry							
Conformation occlusion	21	0.5781	0.5437	0.4881	0.4008	0.2275	0.1124
Crystal periodicity	4	0.6667	0.6667	0.4444	0.6111	0.3333	0.2222
Dihedral measurement	9	0.4259	0.5000	0.3889	0.3889	0.3889	0.4074
Hybridization state	7	0.5238	0.5555	0.5397	0.5238	0.7143	0.6667
Molecular extent	9	0.2963	0.2778	0.2778	0.2778	0.1667	0.0556
Domain mean	50	0.4995	0.4995	0.4361	0.4106	0.3222	0.2417
Mechanical CAD							
Assembly composition count	10	0.0667	0.1000	0.0333	0.1000	0.1000	0.0000
Mechanism motion	6	0.2222	0.3889	0.1111	0.1111	0.0000	0.0000
Part instance count	20	0.0500	0.0667	0.1167	0.0500	0.0667	0.0000
Part property identification	14	0.1905	0.3333	0.1905	0.1429	0.2619	0.2619
Domain mean	50	0.1133	0.1867	0.1200	0.0933	0.1200	0.0733
Medical imaging							
Anatomy pointing	5	0.6667	0.5333	0.4667	0.1333	0.2667	0.0000
Geometry measurement	8	0.4410	0.4757	0.3958	0.2500	0.5104	0.1146
Lesion pointing	25	0.0933	0.0400	0.0000	0.0267	0.0400	0.0267
Slice comparison	12	0.5694	0.5972	0.3472	0.4306	0.4028	0.2083
Domain mean	50	0.3206	0.2928	0.1933	0.1700	0.2250	0.0817
Overall	250	**0.3616**	**0.3609**	**0.2846**	**0.2652**	**0.2435**	**0.1598**

Table 14: Source+MCP subdomain mean task scores, model group 1 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.2708	0.2662	0.2708	0.3125	0.2639	0.2245
Dimensional compliance	16	0.3923	0.3539	0.3081	0.3921	0.3729	0.2025
Interference and clearance	12	0.3130	0.3361	0.3778	0.3384	0.3535	0.2750
Object geometry readings	10	0.8500	0.8167	0.8500	0.7167	0.6333	0.3167
Domain mean	50	0.4356	0.4211	0.4243	0.4250	0.3942	0.2480
Biology							
Interaction typing	8	0.8750	0.8750	0.7500	0.7083	0.1250	0.7500
Molecular contact	14	0.9286	0.9048	0.8810	0.8571	0.7857	0.8810
Occluded target geometry	6	0.0926	0.1111	0.4630	0.0926	0.2778	0.1852
Proximity choice	8	0.5417	0.5000	0.6250	0.5417	0.6250	0.5417
Spatial arrangement	14	0.6429	0.6429	0.6310	0.6071	0.6131	0.5476
Domain mean	50	0.6778	0.6667	0.6989	0.6211	0.5450	0.6289
Chemistry							
Conformation occlusion	21	0.6230	0.6561	0.5886	0.5556	0.6601	0.4841
Crystal periodicity	4	0.8333	0.8056	0.6945	0.6945	0.7084	0.5000
Dihedral measurement	9	0.5000	0.5000	0.4259	0.5000	0.5556	0.4074
Hybridization state	7	0.7460	0.7143	0.7143	0.8413	0.5714	0.4286
Molecular extent	9	0.3210	0.3395	0.2778	0.2778	0.2778	0.2963
Domain mean	50	0.5806	0.5911	0.5295	0.5467	0.5639	0.4300
Mechanical CAD							
Assembly composition count	10	0.4000	0.4000	0.3000	0.1667	0.4167	0.1000
Mechanism motion	6	0.5000	0.5000	0.4444	0.3333	0.3889	0.3333
Part instance count	20	0.6333	0.4500	0.1500	0.0500	0.0750	0.0833
Part property identification	14	0.7619	0.7857	0.6429	0.5000	0.5714	0.4286
Domain mean	50	0.6067	0.5400	0.3533	0.2333	0.3200	0.2133
Medical imaging							
Anatomy pointing	5	0.8000	0.7333	0.5333	0.6000	0.6667	0.6000
Geometry measurement	8	0.5243	0.4965	0.5382	0.5556	0.4410	0.5903
Lesion pointing	25	0.1600	0.1733	0.1333	0.1467	0.1467	0.1333
Slice comparison	12	0.4028	0.4028	0.4306	0.5139	0.5278	0.5278
Domain mean	50	0.3406	0.3361	0.3094	0.3456	0.3372	0.3478
Overall	250	**0.5282**	**0.5110**	**0.4631**	**0.4343**	**0.4321**	**0.3736**

Table 15: Source+MCP subdomain mean task scores, model group 2 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.2500	0.2315	0.2245	0.2060	0.2245	0.1134
Dimensional compliance	16	0.3445	0.3455	0.2512	0.1684	0.2175	0.1033
Interference and clearance	12	0.3778	0.3454	0.1977	0.2000	0.1181	0.0556
Object geometry readings	10	0.3167	0.2667	0.2000	0.0667	0.1167	0.0667
Domain mean	50	0.3242	0.3023	0.2217	0.1647	0.1751	0.0870
Biology							
Interaction typing	8	0.7083	0.5417	0.1250	0.1250	0.0833	0.1250
Molecular contact	14	0.8571	0.8333	0.6905	0.7143	0.7619	0.7619
Occluded target geometry	6	0.1296	0.1852	0.2037	0.0741	0.0926	0.1481
Proximity choice	8	0.5000	0.5417	0.4583	0.5000	0.2083	0.4167
Spatial arrangement	14	0.5476	0.5119	0.5357	0.5119	0.4286	0.3214
Domain mean	50	0.6022	0.5722	0.4611	0.4522	0.3911	0.4078
Chemistry							
Conformation occlusion	21	0.4815	0.4709	0.4312	0.3585	0.3585	0.2050
Crystal periodicity	4	0.6667	0.5556	0.6667	0.6667	0.3889	0.1111
Dihedral measurement	9	0.3148	0.3148	0.4630	0.3889	0.2778	0.4444
Hybridization state	7	0.5238	0.5397	0.6508	0.5873	0.6825	0.4762
Molecular extent	9	0.2963	0.3148	0.2222	0.2778	0.1852	0.1481
Domain mean	50	0.4389	0.4311	0.4489	0.4061	0.3606	0.2683
Mechanical CAD							
Assembly composition count	10	0.0667	0.0333	0.0667	0.0667	0.0000	0.0333
Mechanism motion	6	0.2222	0.2222	0.0000	0.0556	0.1111	0.0000
Part instance count	20	0.1000	0.0500	0.0833	0.0833	0.0500	0.0000
Part property identification	14	0.1667	0.3333	0.0714	0.2619	0.3571	0.1190
Domain mean	50	0.1267	0.1467	0.0667	0.1267	0.1333	0.0400
Medical imaging							
Anatomy pointing	5	0.5333	0.3333	0.3333	0.1333	0.4667	0.0000
Geometry measurement	8	0.4688	0.4306	0.3785	0.1840	0.3750	0.1424
Lesion pointing	25	0.1600	0.0267	0.0400	0.0267	0.0133	0.0000
Slice comparison	12	0.5833	0.4861	0.4861	0.5556	0.3750	0.1806
Domain mean	50	0.3483	0.2322	0.2306	0.1894	0.2033	0.0661
Overall	250	**0.3681**	**0.3369**	**0.2858**	**0.2678**	**0.2527**	**0.1738**

Table 16: Fixed6 subdomain mean task scores, model group 1 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.1528	0.1361	0.1815	0.1894	0.0972	0.1931
Dimensional compliance	16	0.1251	0.1085	0.2211	0.1851	0.0875	0.1172
Interference and clearance	12	0.1806	0.1713	0.2963	0.3009	0.1644	0.2417
Object geometry readings	10	0.1500	0.1500	0.2667	0.2500	0.0500	0.2333
Domain mean	50	0.1500	0.1385	0.2387	0.2269	0.1008	0.1885
Biology							
Interaction typing	8	0.3750	0.2917	0.7500	0.3750	0.1250	0.6667
Molecular contact	14	0.0000	0.0000	0.4524	0.0000	0.0952	0.5000
Occluded target geometry	6	0.0000	0.0000	0.3148	0.0926	0.1481	0.1667
Proximity choice	8	0.4583	0.4167	0.2500	0.4167	0.5833	0.4583
Spatial arrangement	14	0.5952	0.5119	0.5952	0.5714	0.4524	0.4286
Domain mean	50	0.3000	0.2567	0.4911	0.2978	0.2844	0.4600
Chemistry							
Conformation occlusion	21	0.1058	0.0794	0.2883	0.2010	0.0529	0.1468
Crystal periodicity	4	0.0000	0.0000	0.5000	0.5000	0.3889	0.4722
Dihedral measurement	9	0.0741	0.0741	0.0370	0.0000	0.0000	0.0370
Hybridization state	7	0.8413	0.6667	0.7460	0.8095	0.2540	0.5238
Molecular extent	9	0.0000	0.0000	0.1667	0.1111	0.0000	0.0556
Domain mean	50	0.1756	0.1400	0.3022	0.2578	0.0889	0.1894
Mechanical CAD							
Assembly composition count	10	0.4333	0.3000	0.3000	0.2667	0.1333	0.3333
Mechanism motion	6	0.4444	0.3333	0.5000	0.3333	0.2778	0.2778
Part instance count	20	0.0500	0.0500	0.0000	0.0000	0.0000	0.0000
Part property identification	14	0.6429	0.5238	0.2143	0.4286	0.2857	0.3810
Domain mean	50	0.3400	0.2667	0.1800	0.2133	0.1400	0.2067
Medical imaging							
Anatomy pointing	5	0.6667	0.7333	0.7333	0.5333	0.0000	0.1333
Geometry measurement	8	0.0417	0.0417	0.2535	0.1389	0.1250	0.2326
Lesion pointing	25	0.2000	0.1200	0.0533	0.0267	0.0000	0.0000
Slice comparison	12	0.4583	0.3333	0.5417	0.4444	0.4028	0.5417
Domain mean	50	0.2833	0.2200	0.2706	0.1956	0.1167	0.1806
Overall	250	**0.2498**	**0.2044**	**0.2965**	**0.2383**	**0.1462**	**0.2450**

Table 17: Fixed6 subdomain mean task scores, model group 2 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.1486	0.1259	0.0810	0.1343	0.0694	0.0139
Dimensional compliance	16	0.1680	0.1600	0.0568	0.0925	0.0104	0.0000
Interference and clearance	12	0.3083	0.2625	0.1954	0.1574	0.1065	0.0278
Object geometry readings	10	0.1500	0.1167	0.1833	0.0667	0.0167	0.0000
Domain mean	50	0.1934	0.1678	0.1212	0.1129	0.0489	0.0100
Biology							
Interaction typing	8	0.7083	0.3750	0.0417	0.0833	0.0000	0.0833
Molecular contact	14	0.6429	0.5952	0.5000	0.4524	0.3571	0.5714
Occluded target geometry	6	0.2222	0.0926	0.0556	0.0000	0.0000	0.0000
Proximity choice	8	0.4167	0.4583	0.2500	0.2500	0.0833	0.2500
Spatial arrangement	14	0.5714	0.5357	0.5714	0.5595	0.4643	0.4286
Domain mean	50	0.5467	0.4611	0.3533	0.3367	0.2433	0.3333
Chemistry							
Conformation occlusion	21	0.1468	0.0529	0.0370	0.0635	0.0000	0.0000
Crystal periodicity	4	0.4166	0.4167	0.1944	0.1667	0.0556	0.0000
Dihedral measurement	9	0.0185	0.0741	0.0000	0.0000	0.0000	0.0000
Hybridization state	7	0.4921	0.5079	0.4762	0.4762	0.4603	0.0159
Molecular extent	9	0.0185	0.0000	0.0000	0.0000	0.0000	0.0000
Domain mean	50	0.1705	0.1400	0.0978	0.1067	0.0689	0.0022
Mechanical CAD							
Assembly composition count	10	0.3000	0.1333	0.1667	0.3667	0.1333	0.0000
Mechanism motion	6	0.2778	0.4444	0.0000	0.1111	0.1111	0.0000
Part instance count	20	0.0000	0.0000	0.0000	0.0167	0.0000	0.0000
Part property identification	14	0.2143	0.2857	0.2619	0.2143	0.3095	0.1667
Domain mean	50	0.1533	0.1600	0.1067	0.1533	0.1267	0.0467
Medical imaging							
Anatomy pointing	5	0.1333	0.0000	0.0000	0.0000	0.0000	0.0000
Geometry measurement	8	0.1458	0.1562	0.0521	0.0694	0.0000	0.0000
Lesion pointing	25	0.0000	0.0133	0.0000	0.0000	0.0000	0.0000
Slice comparison	12	0.5556	0.4722	0.3056	0.4583	0.2639	0.0833
Domain mean	50	0.1700	0.1450	0.0817	0.1211	0.0633	0.0200
Overall	250	**0.2468**	**0.2148**	**0.1521**	**0.1661**	**0.1102**	**0.0824**

Table 18: Source+MCP minus Source in percentage points. Pointwise percentile intervals use 2,000 paired bootstrap resamples of 250 task IDs (seed 20261029), holding exported repeat means fixed. They describe task resampling conditional on the evaluated outputs, rather than run-to-run variability or a controlled MCP intervention. No multiple-comparison adjustment is applied.

Model	Difference	2.5%	97.5%
GPT-6 Astra	-0.08	-1.94	+1.83
GPT-6.1 Sol	+2.17	+0.34	+4.19
Claude Fable 5.1	+0.12	-1.95	+2.17
Claude Opus 5.5	-1.96	-3.83	-0.23
Gemini 4 Argon	+0.43	-2.74	+3.52
Qwen3.8 Max 0902	-0.94	-3.65	+1.60
Qwen3.8 Flash	+0.64	-1.96	+3.22
Kimi K3	-2.40	-5.15	+0.24
GLM-5.3 Flash	+0.12	-2.31	+2.82
DeepSeek V4 Flash	+0.27	-1.97	+2.62
Doubao Seed Evolving	+0.92	-2.28	+4.14
Hunyuan-4 Preview	+1.41	-1.13	+4.07

Table 19: Descriptive score profile across the 36 canonical configurations. Each task is assigned according to its highest available-repeat mean: zero, strictly between zero and one, or one. A full mean for one configuration does not imply success for every model or individual repeat.

Domain	All zero	Partial maximum	Full-score maximum
Architecture	6	27	17
Biology	4	11	35
Chemistry	6	21	23
Mechanical CAD	5	10	35
Medical imaging	13	14	23

E.4 Paired task differences

We resample 250 task IDs with replacement 2,000 times using seed 20261029. Each draw retains paired Source and Source+MCP means, including their available-repeat handling. The interval endpoints are the 2.5th and 97.5th percentiles, using linear interpolation between ordered bootstrap estimates. The generated estimates reproduce the frozen analysis exactly.

The 12 intervals in Table 18 are pointwise, with no family-wise or false-discovery adjustment. Their interpretation is conditional on the exported outputs and benchmark construction. They do not capture repeated-run variability, model-version changes, or a controlled tool-access effect.

E.5 Task score profile

For each task, we take the highest available-repeat mean across the 36 canonical configurations. Table 19 retains three categories: zero, strictly between zero and one, and one. The totals are 34, 83, and 133 tasks. A maximum of one requires a full-score mean for at least one configuration, rather than one successful repeat or success by all agents.

E.6 CAD target-definition sensitivity

Excluded IDs: `hard-025`, `hard-026`, `hard-028`, `hard-029`, `hard-031`, `hard-032`, `hard-033`, and `hard-036`. The prior audit reproduced seven stored counts as recursively placed solid-body counts. For `hard-026`, the stored seven instead counts distinct immediate child product definitions; the whole-assembly solid-body count is 26. The questions lack stable target identifiers or explicit counting units, so reproducing a stored number does not resolve the intended target. The supplied audit records establish no numerical gold error.

Table 20: Published 250-task means and the 242-task diagnostic excluding eight CAD questions with underspecified targets or counting units. The diagnostic weights 42 CAD and 50 tasks from each other domain equally per task. The exclusion neither changes the published benchmark nor certifies the remaining references.

	Source		Source+MCP		Fixed6	
Model	250	242	250	242	250	242
GPT-6 Astra	0.5291	0.5356	0.5282	0.5292	0.2498	0.2580
GPT-6.1 Sol	0.4893	0.5013	0.5110	0.5183	0.2044	0.2097
Claude Fable 5.1	0.4619	0.4772	0.4631	0.4784	0.2965	0.3063
Claude Opus 5.5	0.4539	0.4689	0.4343	0.4487	0.2383	0.2461
Gemini 4 Argon	0.4278	0.4419	0.4321	0.4463	0.1462	0.1510
Qwen3.8 Max 0902	0.3830	0.3956	0.3736	0.3860	0.2450	0.2531
Qwen3.8 Flash	0.3616	0.3736	0.3681	0.3802	0.2468	0.2550
Kimi K3	0.3609	0.3728	0.3369	0.3481	0.2148	0.2219
GLM-5.3 Flash	0.2846	0.2940	0.2858	0.2952	0.1521	0.1572
DeepSeek V4 Flash	0.2652	0.2739	0.2678	0.2767	0.1661	0.1703
Doubao Seed Evolving	0.2435	0.2516	0.2527	0.2610	0.1102	0.1139
Hunyuan-4 Preview	0.1598	0.1650	0.1738	0.1796	0.0824	0.0852

The diagnostic removes these eight IDs and averages the remaining 242 tasks equally: 42 CAD and 50 from every other domain. It preserves the complete 12-model ranking under every condition, both source-over-Fixed6 directions for every model, and all 12 MCP difference signs. The largest mean change is 1.531 percentage points for Fable Source+MCP. The original task membership, questions, references, and scores remain unchanged.

The frozen supplement includes the prior audit report, independent count outputs, input checksums, and count-reader source. Regenerating this score diagnostic uses the frozen exported scores; it does not rerun that structural audit, which additionally requires the pinned STEP assets.

E.7 Scope of the execution-configuration comparison

The two cross-harness tables appear together in Section 6.3.1. Each cell first averages the three exported scores within a task and then averages tasks equally. The 250-task and 94-task comparisons remain separate because their cohorts and source layers differ. No paired significance test or causal harness estimate is inferred from these cross-batch contrasts.

F Historical Cohort Supplements

F.1 Eight-arm camera sweep

Section 6.3.2 reports the historical camera takeaway and compact mean table. Table 21 preserves the complete per-arm diagnostics and separate current Fixed6 anchors. The source package contains 250 v11 tasks and nine score columns: eight camera arms and the MCP-only arm discussed in Section 6.3.3, with 250 values per column. The current release provides a 12-model Fixed6 anchor, but no version-matched camera-plan sweep. Historical documentation reports $2N$ images per medical task, with half identical between aligned and orbit plans. This limits the interpretation of medical camera-plan differences; the original render manifests are unavailable for independent rechecking.

Table 21: Twelve views do not improve either historical camera plan over six. Historical v11 camera sweep and current Fixed6 anchors. Panel A contains eight camera arms; the source package's ninth arm is MCP-only (Table 5). The panels use different cohorts, execution conditions and repeat policies and must not be differenced.

Panel A: historical v11, GPT-6 Astra with Claude Code + LiteLLM

Plan	N	Mean	Full	Zero	Ineligible	Retried
Aligned	1	0.1223	24	210	8	10
Aligned	3	0.1516	29	198	8	9
Aligned	6	0.1529	29	198	8	6
Aligned	12	0.1427	26	199	8	2
Orbit	1	0.1171	23	211	8	3
Orbit	3	0.1204	22	207	8	5
Orbit	6	0.1463	27	199	8	4
Orbit	12	0.1463	28	199	8	2

Panel B: selected entries from the current 12-model Fixed6 release

Model	Mean task score
GPT-6 Astra	0.2498
GPT-6.1 Sol	0.2044
Claude Fable 5.1	0.2965
Claude Opus 5.5	0.2383

Panel A (S1): 250 tasks per arm, v11 cohort; max effort, last attempt retained, host-side `Sci3dVerifier`. Its intersection with final250 is the 205 retained IDs; the 32 replacement and 13 later-replacement IDs are absent. Question, gold, scorer and camera manifests are not locked to the current bundle. Full and Zero count tasks whose retained exported score is one or zero. Ineligible and Retried are task counts copied from the per-arm diagnostics; they can overlap Zero and do not add repeats or remove tasks from the denominator. The historical documentation reports $2N$ image files for each medical task and half of those inputs identical between aligned and orbit plans. Complete run trees and render manifests are absent, so these input declarations are not independently reverified here. **Panel B (S2):** route `cc2d-6`, 12 models $\times$ 250 tasks, three scores per task for 11 models; Hunyuan has 247 tasks with three scores and three tasks with two scores. The export does not establish per-row Fixed6 harness metadata. The selected current Astra 3D configuration uses Codex, whereas the historical Astra sweep used Claude Code + LiteLLM; matching the model name does not align the runs. The release supplies only an $N = 6$ anchor, with no current $N = 1, 3, 12$ or orbit comparison. Neither panel establishes that Fixed6 is optimal for the current benchmark.

F.2 Historical MCP-only record

Section 6.3.3 and Table 5 report the MCP-only access distinction in the main text. Its source is `v11-250-2d-9arm-20261009/results/per-task.csv`, with an exported mean of 0.1756667 over 250 tasks. The historical README declares a raw-asset MCP-only tool surface, but the local package contains no tool list, disabled Bash/Read/Write parameters, environment record or trajectories. File and tool isolation remain unverified. This is a different access protocol from the current Source+MCP route, which retains source data and code execution; no same-version MCP-only pair is available.